

Kuiper TTS: Making Legacy Scientific Papers Accessible by Converting Equations into Spoken Language

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Abstract - Our project aims to create a tool that is easy for any person to use to enjoy the beauty of mathematics and science. Scientific papers are often inaccessible to low vision readers because legacy PDFs expose little semantic structure for screen readers, and the problem is amplified by mathematical and scientific notation. For many other readers, challenges such as dyslexia, difficulty in learning symbols, or having a limited mathematical background also create a barrier to learning. Meanwhile, standard PDF-to-text or Optical Character Recognition (OCR) tools available often omit equations or reduce them to unreadable fragments. Current screen-readers are extremely difficult to use, requiring expertise, and are unreliable at reading equations. We present a user-focused system that shifts the focus from screen reader to an audio player capable of reading an ordinary text file and reading math and scientific content aloud with high fidelity. In this paper, we describe a method to prepare a LaTeX document in a batch from PDF in advance and other sources, mindfully reducing power consumption. The audio player will be capable of functioning for hours on a small platform such as a cell phone, will not require an internet connection, and will be freely available and open source.

Index Terms - Mathematics Education, Text-to-Speech, Accessibility, Math-to-Speech, LaTeX, MathML, LLM, PDF, OCR, Optical Character Recognition.

INTRODUCTION

The body of human knowledge – particularly in technical fields such as mathematics and engineering – resides largely within curated repositories of formal literature: digital archives, scholarly libraries, and structured electronic publications. Particularly in technical topics, most documents are formatted in Portable Document Format (PDF) and legacy formats. PDF provides a single visual format portable across platforms, but is not well-designed for the purpose of disseminating equations and other artifacts, nor can it be easily read by a screen reader. There are also documents remaining in print and handwriting that remain unscanned. Newer scientific papers are often in either LaTeX or Word, preserving more of the semantic content. Our Kuiper-TTS project includes a software

component to read a text file as audio, and a set of agents and AI processors that convert legacy scientific papers into an intermediate format where the equations, figures, and images have been validated, and spoken equivalents pre-computed. The result is a document that can be played in audio very efficiently. Because the presentation of equations in PDF can be ambiguous, preprocessing with multiple agents and AI filters is used to try to preserve the semantic content and store it in a precise form. We use LaTeX as the intermediate form, embedding any content for the audio player in tags that identify how it is to be handled. The result is an audio player that can read a normal text file, but can also read a scientific paper with equations. The benefit is to provide for low-vision users, a hands-free application (such as driving) or an augmented reading experience for people with dyslexia, and for validating input to ensure AI imports the content correctly for retrieval-augmented generation (RAG). Our experiments show that many AIs trained to perform OCR on PDF files will fail to read equations at the top and bottom of the page because they must reject erroneous header and, for example, footer text. Because of this, it seems likely that many AIs are being trained on incomplete information. The pipeline we have defined to read, edit, and correct PDF files and save the results may potentially increase the accuracy of LLMs trained from scientific papers, though this has not yet been tested. Scientific knowledge is overwhelmingly distributed as PDFs, including decades of archival proceedings, technical reports, and scanned journals [1-5]. While PDFs preserve visual layout, they expose weak semantics: screen readers often follow a flawed linear order, drop structure, and render mathematics as fragmented or unusable for text-to-speech. This also means that AI OCR correction is imperfect.

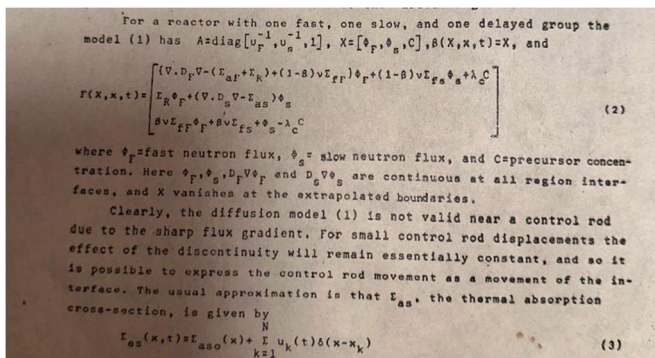


FIGURE I

LEGACY PAPER NON-IDEAL OPTICS

Equations can be garbled, discarded, or even randomly hallucinated. As a result, legacy scientific papers remain hard to consume for blind and low-vision readers and can also burden readers with dyslexia. There are significant reliability challenges with OCR. Figure I shows a paper that is scanned using a cell phone. There is shadowing, pages are curved, and even discoloration due to spilled liquids. Figure II shows an online page from Turing's original submissions. Of course, for this paper, it is famous enough that it is probably also available online, translated by hand. Generally, however, this is not guaranteed.

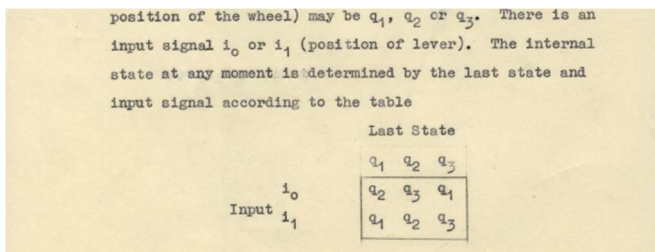


FIGURE II

DIAGRAM FROM TURING'S ORIGINAL PAPER

Figure III shows a section of the bibliography from the Turing paper. The section displays overstrike errors and extra lines. Imperfections can be very challenging.

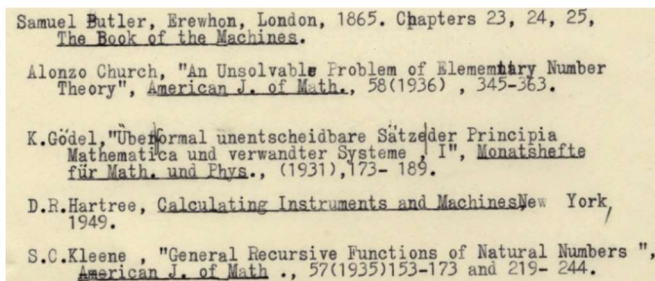


FIGURE III

TURING BIBLIOGRAPHY SNIPPET

Figure IV is from a Daubechies paper on waveforms. The artifact includes a lot of noise.

We use the following convention for the Fourier transform (in one dimension):

$$(Ff)(\xi) = \hat{f}(\xi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx e^{-ix\xi} f(x). \quad (0.0.1)$$

With this normalization, one has

$$\|\hat{f}\|_{L^2} = \|f\|_{L^2},$$

$$|\hat{f}(\xi)| \leq (2\pi)^{-1/2} \|f\|_{L^1},$$

where

$$\|f\|_{L^p} = \left[\int_{\mathbb{R}} dx |f(x)|^p \right]^{1/p}. \quad (0.0.2)$$

FIGURE IV

DAUBECHIES SHOWING SCANNING ARTIFACTS

Many papers are available in PDF in better condition. Most new papers are available on arxiv.org, meaning that all new papers are already in LaTeX [8-14]. The same missing structure can mislead AI systems that ingest raw PDF text, motivating conversion into an interpretable form that humans can audit and assistive tools can reliably traverse. This reframes accessibility as a joint human-AI parsing problem. We improve text-to-speech (TTS) fidelity by recovering document semantics prior to narration. Representing mathematics and structure in a formal language (e.g., LaTeX, or Markdown with embedded LaTeX/MathML) makes equation boundaries, grouping, and operator hierarchy explicit, enabling robust math-aware speech and navigation. Accordingly, we use PDF-to-LaTeX not as an end goal, but as a structured intermediate representation that reduces omissions and improves traversability. [15-19]. We present an accessibility-oriented pipeline that converts legacy scientific PDFs into a screen-reader-friendly representation by generating a constrained LaTeX subset that encodes a non-visual document model. The output preserves prose, mathematical structure, and the metadata required for audio rendering and navigable traversal, enabling speech-first consumption with substantially higher fidelity than direct PDF playback. Our goal is automatic conversion from PDF into a structured intermediate representation that preserves equations and document organization, supporting accurate math-aware TTS and reliable navigation. In this first prototype, we are focused only on equations, which is a large, complex topic. For a full solution, however, eventually alternative forms of other artifacts, including images/diagrams/figures, tables, citations, and other layout-based features, will have to be encoded in audio.

ARCHITECTURE

To read technical documents aloud, a TTS engine is needed to convert English text (and, in the future, other languages) to speech, and a step to capture the existing paper and convert it to an intermediate format that is unambiguous. As shown in Fig. V, the workflow comprises preprocessing (PDF ingestion and page rendering), per-page conversion via a vision-language model (VLM) invoked through the Hugging Face API, post-processing (content cleaning), assembly (page combination and document generation), and

output stages (validation and final .tex generation), with optional figure/image extraction. For PDF papers, we implemented two converters. The first uses *deepseek_ocr* and attempts to convert a PDF to Markdown. This engine is running on our local desktop with 32GB RAM and 12GB of video RAM on an NVIDIA 3060 and can convert a sample 7-page paper in a few minutes, but makes significant errors, rendering it unusable for now. The second attempt was more successful, using Qwen2.5-VL-7B-Instruct [57]. While working with the text extraction, it became obvious that the very first step of our pipeline is critically important; therefore, we explored the best practices of OCR and their suitability for our project.

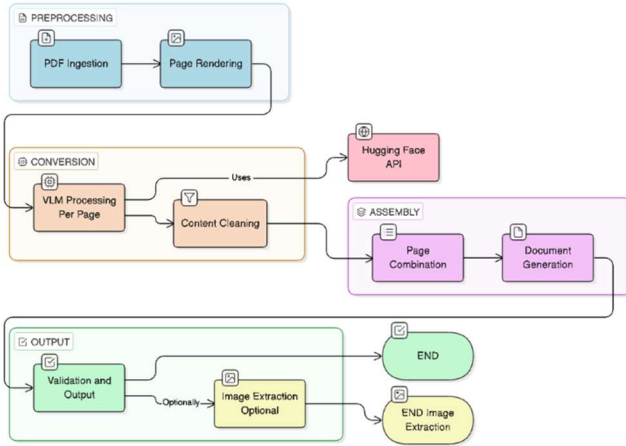


FIGURE V

BATCH PDF-TO-ACCESSIBLE-TEXT PIPELINE WITH PAGE RENDERING, VLM-BASED EXTRACTION, CLEANING, ASSEMBLY, AND OPTIONAL IMAGE EXTRACTION.

I. Top OCR Tools and Vision-Language Models

Our exploration of OCR discovered nearly 100 tools capable of OCR. Figure VI demonstrates a summary returned by GPT-4o via API after showing it Example 1 (Fig. I).

```
Result: The image contains a section of a technical document
discussing a mathematical model for a reactor with one fast
, one slow, and one delayed group. It describes parameters a
nd equations related to neutron flux and precursor concentra
tion.

Key elements include:

- A matrix equation involving terms like  $\nabla$ ,  $D_F$ ,  $\Sigma$ , and  $\lambda$ , which relate to fast
neutron flux  $\Phi_F$  and slow neutron flux  $\Phi_S$ .
- A passage explaining that the diffusion model is not valid
near control rods due to sharp flux gradients.
- Variables include:
  -  $\Phi_F$  = fast neutron flux
  -  $\Phi_S$  = slow neutron flux
  -  $C$  = precursor concentration
  -  $\Sigma_{as}$  = thermal absorption cross-section
- There's mention of boundary conditions and continuity at r
egion interfaces.
- An equation labeled (3) provides an expression for  $\Sigma_{ma_{as}}(x,t)$ .

The content is highly specialized, likely from a nuclear phy
sics or engineering context.
```

FIGURE VI

GPT-4o ANALYSIS OF EXAMPLE 1.

Understandably, it is natural to further exploit the OpenAI API and continue a conversation with the chatbot regarding the provided response, making it an AI tutor. We prefer working with open-source tools but require high accuracy in formula preservation and clarity (see Table I). Two main approaches for text rendering - a stable Python pipeline via custom code or library, and a multimodal AI model like Gemini, ChatGPT, Claude, Mistral, DeepSeek, Ernie, Hunyuan, or Qwen that would understand the context and interpret the formulas [21]. It is important to mention that the various versions of the same models can provide significantly different results, and the process of their improvement is, as of now, unstoppable.

TABLE I

OCR AND AUXILIARY EXTRACTION TOOLS, CAPABLE OF FORMULA EXTRACTIONS.

#	Ref	Tool	Role	Type	Free	OSS	Eq
1	[27]	OCR-D	OCR	Suite	Yes	Yes	M
2	[28]	Surya OCR	OCR	Toolkit	Yes	Yes	M
3	[29]	olmOCR	PDF	Toolkit	Yes	Yes	M
4	[30]	Mistral OCR	OCR	API	No*	No	M
5	[31]	Mathpix	Eq	App/API	No	No	H
6	[32]	Nougat	PDF	Model	Yes	Yes	H
7	[33]	GROBID	Parse	Parser	Yes	Yes	M
8	[34]	Marker	PDF	Toolkit	Yes	Yes	M
9	[35]	pix2tex	Eq	Model	Yes	Yes	H
10	[36]	Pix2Text	Eq	Toolkit	Yes	Yes	H
11	[37]	ABBYY FineReader	OCR	Desktop	No	No	M
12	[38]	Adobe PDF Services API	PDF	API	No	No	M
13	[39]	Docling	PDF	Converter	Yes	Yes	M
14	[40]	LayoutParser	Layout	Framework	Yes	Yes	M
15	[41]	deeptext	Layout	Pipeline	Yes	Yes	M
16	[42]	Donut	Layout	Model	Yes	Yes	M
17	[43]	LayoutLMv3	Layout	Model	Yes	Yes	M
18	[44]	DiT	Layout	Model	Yes	Yes	M
19	[45]	GPT-4o (Vision)	VLM	VLM	No*	No	M
20	[46]	Claude (Vision)	VLM	VLM	No*	No	M
21	[47]	Gemini (Vision)	VLM	VLM	No*	No	M
22	[48]	Grok (Vision)	VLM	VLM	No*	No	M
23	[49]	DeepSeek (VLM/LLM)	VLM	Model	Yes	Yes	M
24	[20]	Qwen2.5-VL	VLM	Model	Yes	Yes	M
25	[50]	InternVL	VLM	Model	Yes	Yes	M

Eq legend: H=equation-focused/high formula fidelity. M=scientific/layout aware. *Paid API (may offer trials).

METHODOLOGY

We first render each PDF page to an RGB raster image using PyMuPDF [52]. PyMuPDF exposes page rendering through *Page.get_pixmap()*, with resolution controlled either via an explicit transformation matrix or directly through a DPI parameter (e.g., dpi=300)[52]. Rendering is performed page-by-page rather than on the entire document at once. This decision is driven by (i) robustness to long documents that can exceed practical context limits, (ii) improved fault isolation (a single problematic page does not poison the full conversion), and (iii) the ability to tune resolution per page when small fonts or dense equations require higher fidelity.

A) VLM Transcription with Structure-Aware Prompting

For each rendered page, we invoke a VLM with an instruction prompt that biases the model toward emitting LaTeX rather than prose. The prompt is context-aware: the first pages emphasize title/author/abstract blocks, as well as stable document scaffolding. The body pages emphasize section structure, display math, tables, and floats. The final pages emphasize references and appendices. The goal is not to perfectly replicate the PDF’s typography, but to produce a semantically faithful and compilable LaTeX document that can be refined downstream. When calling hosted inference, we use the Hugging Face Inference interfaces for VLMs, which support passing images either as URLs or base64-encoded data in the prompt stream [22].

B) Model Suite and Routing

At this point, we use Qwen2.5-VL-7B-Instruct as the default model because it is explicitly positioned as a strong general-purpose VLM and reports strong performance on document-centric benchmarks (e.g., DocVQA, ChartQA, OCRBench) in its released evaluations [23]. In practice, however, PDF pages are heterogeneous, so we support routing to specialized models when the page content warrants it. For math and scientific-notation-heavy pages, we optionally route to nanonets/Nanonets-OCR-s, which is designed for structured document transcription and explicitly advertises LaTeX equation recognition as part of its feature set [24]. For throughput-oriented settings where speed and memory are prioritized over maximum fidelity, we optionally route to reducto/RolmOCR, which is described as a faster, lower-memory alternative while still targeting document OCR performance [25]. For pages where formatted OCR behavior is particularly important, we optionally use stepfun-ai/GOT-OCR2_0, which is an OCR model that spans plain, formatted, table, chart, and formula-like content [26]. This routing is implemented as a policy decision (e.g., configuration or heuristics such as equation density), allowing the system to trade off accuracy, latency, and cost without changing the surrounding pipeline.

C) LaTeX Cleaning and Validation for Compilability

Raw model output is treated as a best-effort draft rather than a guaranteed compilable artifact. We therefore apply deterministic cleanup that removes common wrappers (e.g., markdown fences), normalizes escaping and whitespace, and repairs frequent syntactic issues such as unbalanced braces or unclosed environments. We then validate structural correctness through brace and environment matching and optionally run a compilation test when a LaTeX toolchain is available. This stage is essential because even strong VLMs occasionally emit near-correct LaTeX that fail due to minor syntax errors; deterministic repair converts some of these failures into compilable outputs while providing clear failure signals when the output is genuinely inconsistent.

D) Cloud Inference and Local Execution with Ollama

The same pipeline supports both hosted and local inference backends. For hosted inference, we call VLM endpoints

through Hugging Face tooling that supports image inputs (including base64) in a standardized client workflow [22]. For fully local execution, we can run multimodal vision models via Ollama. Ollama provides a local HTTP API served by default at <http://localhost:11434/api> and explicitly supports multimodal (vision) models via its multi-modal engine, including Qwen 2.5 VL. In particular, the Ollama model library includes qwen2.5vl as a vision-language model option. Additionally, some OCR/VLM projects distribute an Ollama Modelfile (e.g., Nanonets-OCR-s), enabling local packaging and inference through the same Ollama interface. The motivation for local execution is straightforward: it removes rate limits and external dependencies, reduces per-page marginal cost at scale, and supports privacy-sensitive documents, at the expense of local compute and memory requirements.

E) LaTeX Subset Intermediate Format

To capture the semantic content of the paper, the TTS reader requires a document containing information about the paper’s regions (for traversal), diagrams and figures (so they can be described in language rather than drawn), equations, and citations. Currently, we have only implemented equations, though the first-stage conversion does capture section names. We have not attempted to capture images yet. The following shows the LaTeX subset features used in the intermediate document:

```
\section{Introduction}
\cite{Knuth1982}
this \e{word} is emphasized.

\begin{figure}[label]
\desc{this text describes what is in the
      figure, NOT that it IS a figure!}
\end{figure}

\subsection{Background}
\[
x^2 + y^2 = z^2
\eqtext{x squared plus y squared equals
      z squared}
\]
```

F) Text-to-Speech (TTS)

The TTS engine reads the intermediate LaTeX subset and currently only converts text and equations to speech. In the future, a description of diagrams would need to be added, as would the ability to traverse the entire document and its tables rather than reading them.

G) Speech Synthesis Backend: eSpeak NG and Piper

After PDF-to-LaTeX conversion and semantic normalization, we render the recovered text and math narration using a fully local TTS backend (Fig. V). We use eSpeak NG as the grapheme-to-phoneme (G2P) front-end to

obtain consistent phoneme sequences across technical vocabulary and symbols, leveraging its compact, open-source design and broad language/accent-coverage [53]. To synthesize higher-quality waveforms from the resulting phoneme stream, we use Piper, a fast on-device neural TTS engine that embeds eSpeak NG for phonemization, making it well-suited for offline, accessibility- oriented deployment [54]. To reduce latency for long documents with repeated terminology (variables, units, common words), we include a word-level audio cache: cache hits bypass both phonemization and synthesis, while cache misses follow the eSpeak NG $\rightarrow$ Piper path before optional prosody/emphasis post-processing and final audio assembly.

2) Structure-Aware Equation Verbalization

A direct symbol-by-symbol reading of recovered LaTeX is often mathematically correct but perceptually unintelligible in audio form, especially for derivatives, chained equalities, and long expressions with nested structure. This is not primarily a limitation of waveform quality; rather, it is a consequence of losing semantic grouping when mathematical operators are spoken as isolated tokens (e.g., reading dq/dt as “d q over d t” instead of as a single derivative operator).

To address this, we introduce a deterministic, structure-aware narration layer that rewrites equation content into a constrained natural-language form prior to TTS synthesis. This layer does not attempt to control neural prosody directly (Piper does not expose fine-grained prosody controls); instead, it encodes grouping and operator hierarchy explicitly through standardized phrasing and strategically placed pauses. The approach is reproducible and model-agnostic: any TTS system can consume the resulting narration. Key rewrite rules include:

- Derivatives as operators: dq/dt is rendered as “*the derivative of q with respect to t,*” and dv/dt as “*the derivative of v with respect to t,*” preserving the concept of differentiation as a single operator rather than a sequence of letters.
- Partial and mixed partial derivatives: expressions such as $\partial^2 f / \partial x \partial t$ are verbalized as “*the second mixed partial derivative of f with respect to x and t,*” reducing ambiguity in multi-variable contexts.
- Fractions and ratios: simple fractions a/b are typically rendered as “*a over b*” or “*a divided by b,*” but complex numerators/denominators trigger explicit grouping such as “*the quantity . . . over the quantity . . .*”.
- Chained equalities: long chains (e.g., $A = B = C = \dots$) are segmented into clause-level units with short pauses at equality boundaries so the listener can track transformations stepwise.
- Constants, subscripts, and functions: common patterns such as V_0 , $\sin(2\pi ft)$, and ϵ are verbalized consistently (e.g., “*V sub zero*”, “*sine of 2 pi f t*”) to reduce pronunciation drift across the document.

In practice, these rules are applied by parsing (or heuristically tokenizing) the LaTeX math spans and

generating a corresponding narration string (stored alongside the original math). This produces an “audio-first” representation in which mathematical semantics are preserved as language, enabling improved listener retention even with a general-purpose voice model. Future work will fine-tune a neural voice specifically on mathematical narration, which should further improve pacing, emphasis, and naturalness, but the current results already show that semantic grouping in text is a dominant factor in intelligible math audio.

RELATED WORK

Current math-to-speech behavior in the common screen reader and PDF narration pipelines is often inadequate for technical documents, particularly when equations are rendered as symbol sequences rather than semantic operators. Additionally, not all symbols are equivalently stored. For example, in older papers, equations may be hand-drawn artwork and completely unknown, while symbols such as a caret (^) may be spoken. For example, the following equation:

$$\frac{a}{b} + x^3$$

would be spoken as “*a b plus x caret 3*” because the horizontal bar of the fraction is not a character (it is a drawn line) and the caret is only understood as a symbol, so its mathematical meaning is not spoken. Similarly, the order will be wrong for PDF files, for example, with 2 columns or with footnotes. For example:

```
\author{Harald T. Friis...
\section{Capacitors}...
\[
\frac{q}{v} = \text{constant} = C
\]
\eqid{ee.capacitance}
\eqcaption{The ratio of total charge q on
each plate to the voltage v between
the plates is constant}
\eqverbose{The equation fraction numerator
q denominator v equals constant
equals capital C}
\eqterse{q divided by v equals a constant,
C}
```

A) Math-Aware TTS in Prior Accessibility Pipelines

Prior work and mainstream assistive technologies often treat mathematical expressions as either (i) images requiring OCR, (ii) visually formatted text linearized into a token stream, or (iii) markup (e.g., MathML/LaTeX) rendered without strong semantic verbalization rules. In practice, this frequently yields speech that is syntactically faithful but semantically opaque: symbol-by-symbol narration does not reliably communicate operator hierarchy, grouping, or binding, and it overloads working memory for longer expressions. Derivatives, nested fractions, and chained

equalities are especially problematic because they function as operators over structured subexpressions rather than sequences of characters. This motivates a key design choice in our pipeline: rather than relying on a TTS engine to infer mathematical intent from raw tokens, we explicitly generate a constrained natural-language narration that encodes grouping and semantic operators prior to synthesis. This approach is model-agnostic and allows for human intervention if necessary. The file is saved, and the front-end screen reader then plays the resulting text with extremely low computational overhead. Improving voice quality and prosody enhances naturalness, but Intelligibility for technical math is primarily determined by whether the semantic structure is made explicit in the narration text. We currently provide two levels of verbal equation: a verbose beginner, encoded as `\eqverbose`, and a terser `\eqterse`. We generate a unique name for each equation for use in auto-generating an eventual dictionary equivalent to a man-page for equations. Example audio can be accessed on GitHub [57].

TABLE II

OCR BENCHMARK RESULTS (SUCCESSFUL RUNS ONLY).

Paper	Ref	OCR	Chars/Page	Time/Page (s)
Friis	[51]	Qwen2.5 [20]	3703	22
LoRA	[55]	Qwen2.5 [20]	2863	12
BERT	[56]	Qwen2.5 [20]	3488	50

RESULTS

The *deepseek-ocr* version incurs substantial hallucinations, so we are not considering it for now. Our Qwen implementation works well, though it tends to miss equations at the top and bottom of pages. Both implementations are not only open source but can run on an ordinary computer, not even requiring a GPU. This democratizes the ability to batch convert papers to make them accessible. The screen reader will be completely local and open source. For equation conversions, we used Qwen2.5-7B with a prompt prepared by ChatGPT5.2, ChatGPT itself, and the cursor edit agent using the Claude AI. Claude and ChatGPT work far better than smaller LLMs with only 7 billion parameters, but all generate reasonable output. Qwen requires constant supervision to prevent it from injecting pieces of the original equation rather than the spoken equivalent. All models have been improving over time, and our pipeline is designed to benefit from these improvements. Accordingly, an agent will have to be written to supervise the AI and make sure that it performs the desired translation on each equation. Once the equations are transformed (and humans can edit the results, if necessary, though we do not want to need that), we demonstrated that we could read the resulting equations in both *espeak-ng* and *Piper*. We focus on *Piper*. The resulting *tex* file format, which we call *.sumtex* (Summary Tex), stores the corrected equations converted to spoken language and stores a unique name for each equation, which can be used to reference an automatically generated dictionary. The *.sumtex* file can be displayed and read without AI with low power. This preserves the intelligence produced by the AI in a batch

without having to run the AI each time. The pipeline recovers readable prose and structured math environments, producing LaTeX that is largely faithful to the source layout and notation. While occasional formatting and symbol-level errors remain (especially in equation-dense regions), the output is substantially more suitable for downstream parsing and math-aware TTS than direct PDF text extraction.

A) Qualitative Evaluation of Equation Audio Readability

We qualitatively evaluated equation audio by comparing (i) naive symbol-level narration and (ii) the proposed structure-aware verbalization strategy. For short equations, both approaches can be intelligible; however, for longer expressions involving derivatives, chained equalities, or multi-variable notation, symbol-level reading becomes difficult to follow even when the underlying transcription is correct. The structure-aware approach improves listener retention by preserving semantic groupings. Rendering dq/dt as "the derivative of q with respect to t " is consistently easier to parse than "d q over d t," and clause-level segmentation of chained equalities reduces listener overload during multi-step derivations. While the base TTS model still limits natural prosody and emphasis, these results indicate that textual semantic structuring alone can substantially improve mathematical audio accessibility, even without direct prosody control.

B) Equation-Level Accuracy and Error Rate

We evaluate mathematical conversion quality at the level of *numbered equations* in the source PDF, including lettered variants (e.g., 3a, 5a, 6a, 13a, 14a). Each ground-truth equation is labeled as: **Correct** if it appears in the converted LaTeX and matches the PDF mathematically, **Incorrect** if it appears but contains a mathematical/-content error (e.g., wrong coefficient, sign, exponent, or sub-/superscript) or **missing** if it does not appear in the LaTeX output. Across the document, we identify $N = 29$ ground-truth equations, of which $N_c = 23$ are correct, $N_i = 1$ are incorrect, and $N_m = 5$ are missing. The following metrics summarize performance:

$$A \triangleq \frac{N_c}{N} = \frac{23}{29} = 0.7931 \text{ (79.31\%)} \quad (1)$$

$$E \triangleq \frac{N_i + N_m}{N} = \frac{6}{29} = 0.2069 \text{ (20.69\%)} \quad (2)$$

$$C \triangleq \frac{N - N_m}{N} = \frac{24}{29} = 0.8276 \text{ (82.76\%)} \quad (3)$$

$$E | \text{present} \triangleq \frac{N_i}{N - N_m} = \frac{1}{24} = 0.0417 \text{ (4.17\%)} \quad (4)$$

Overall, the dominant failure mode is **omission** (missing equations), while in-expression mathematical errors are comparatively rare among equations that are present in the converted output.

Inductors

Figure 1 shows a long coil of wire with direct current i . The resistance of the wire is assumed to be negligible. By means of a magnetic needle indicator it has been found that the magnetic field H inside the coil is constant. Physicists also found for a closed path around currents that the sum of the magnetic fields along the path multiplied by the path lengths is equal to the total current through the area enclosed by the path. Path $ABCD$ in Fig. 1 shows such a path. The magnetic field is zero along this path except along AD , and the total current through the area of the path is $i \times n$. Thus,

$$Hl = in$$

or, by definition,

$$H = \frac{in}{l} \text{ amperes per meter}$$

In Fig. 2 an ac voltage $e = V_s \cos 2\pi ft$ is applied to the coil; f is the frequency in hertz and t is the time in seconds. (See Fig. 3 for variation of e vs. $2\pi ft$.) Faraday found that the voltage induced in one turn by i is proportional to the area s of the turn and the rate of change of i or H . Hence,

$$e \text{ (one turn)} = \mu s \frac{dH}{dt} = \mu s \frac{n}{l} \frac{di}{dt}$$

where the proportionality factor μ is the permeability of the medium inside the coil (for air, $\mu = 4\pi \times 10^{-7}$ H/m). The voltages in the separate turns are aiding. Therefore, the voltage induced across the coil is

Editor's note: Last month an autobiographical memoir by Dr. Friis, *Seventy-Five Years in an Exciting World*, was published by San Francisco Press, 555 Howard St., San Francisco, Calif.; it also contains "The Wisdom of Harold T. Friis," based on a famous speech (annotated by John R. Pierce) in which the author distills a lifetime's hard-won research experience.

FIGURE VII

CONVERTED LATEX EQUATIONS (VERBATIM)

```
Hl = in; H = \frac{in}{l}; v =
V_0\cos(2\pi f t); v_{\text{one
turn}} = \mu s \frac{dH}{dt} = \mu
s \frac{n}{l}\frac{di}{dt}; \mu =
4\pi \times 10^{-7}, \mathrm{H/m}.
```

The fact that our OCR agent is missing equations highlights a key vulnerability of RAG in general. For a RAG that is reading a paper with many artifacts (images, equations, tables), it is probable that the RAG is ingesting incorrect information (missing, wrong), so this approach is not only useful for accessibility, but also to validate the input data going to AI decision-making across a wide range of fields.

CONCLUSION AND FUTURE WORK

We can use LLMs to quickly convert legacy papers to a format that can be easily read by a screen reader without an LLM. We will present this at the conference. Preprocessing can be done in a batch pipeline once, and the resulting papers can be ready for small, low-power devices, including cellphones. Future work includes a pipeline with multiple levels of editing and validation so that the errors can be reduced. Because we are producing a .tex file, it will be possible for humans to hand-edit the result, if necessary, but we anticipate that this will become increasingly rare. Because our pipeline works with arXiv, it will be possible to validate it by reverse engineering PDF files and comparing the results to the .tex originals. Page-wise conversion can introduce cross-page consistency issues (e.g., repeated macro definitions, inconsistent global numbering, and reference normalization). Future work includes document-level reconciliation passes that synthesize shared macros, harmonize figure/table numbering, and improve bibliography extraction. Another practical limitation is that scanned or degraded PDFs can still challenge VLMs; adaptive DPI selection and more aggressive model routing may mitigate these cases, and domain-specific fine-tuning remains a promising path for higher fidelity in academic papers. We have therefore demonstrated that it should be possible to automate most of the effort required to convert all scientific papers to an accessible format, and to do so without charge or cloud-based spying on the user. Knowledge should be freely available to all. Our initiative is directed to IEEE Standards including: DIITA Transparent

Design for Well-being; P2998 Recommended Practice for Building Accessible Applications[58]; P3337 Standard for Requirements for Designing Accessible Intelligent Cyber-Physical Systems (AICS) for Well-Being[59]; P3386 Standard for Defining and Inferring User Accessibility Needs for Applications including Augmented Reality and Artificial Intelligence Systems [60]; P7020 Standard for Transparency, Accuracy, and Source Integrity in Media and Public-facing Information [61], and others. Providing an open-source, free tool to read files aloud, including scientific information, will be useful in maximizing all human potential to engage in STEM.

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